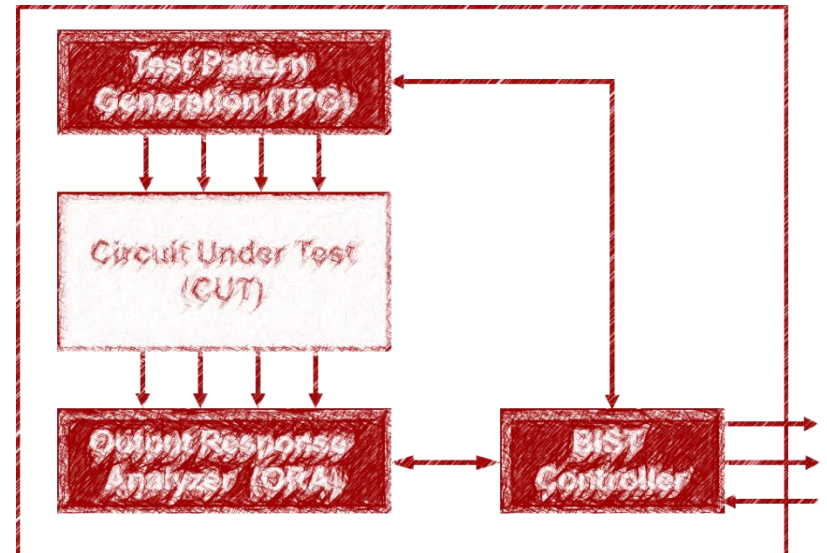


# BIST Part1 - TPG

- Introduction
- Test Pattern Generation
  - ◆ Deterministic: ROM, Algorithm, Counter
  - ◆ Pseudo Random
    - \* Linear Feedback Shift Register, LFSR (1977)
    - \* Cellular Automata, CA (1984)



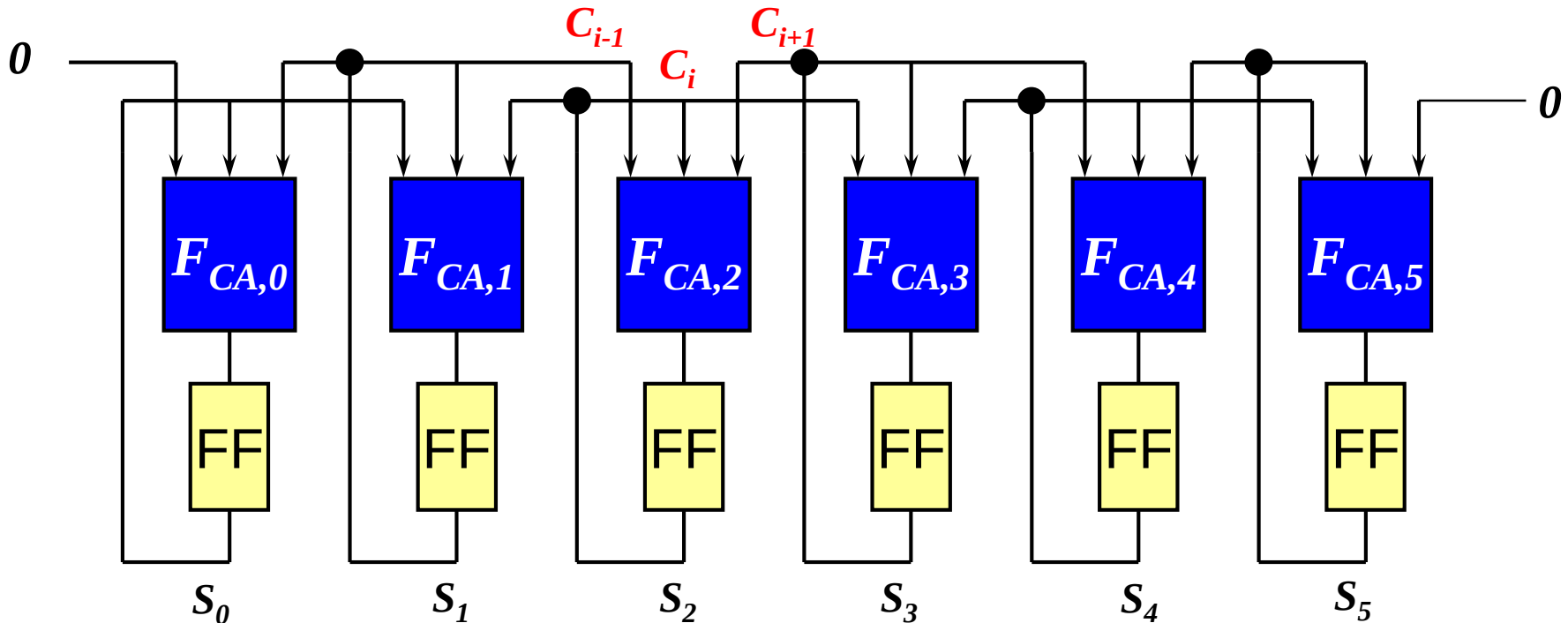
# Cellular Automata [Wolfram 84]

- $F_{CA}$  is a linear function of its two immediate neighbors and itself

$$F_{CA,i} = C_{i-1}S_{i-1} \oplus C_i S_i \oplus C_{i+1}S_{i+1}$$

$C_i=1$  means connection exists

- **Null Boundary Condition** = both ends tied to zero
- **Degree** = number of FF = 6



**CA Has No Long Global Wires**

# Name of CA Cell

- CA cell is named by  $F_{CA}$  function

K map of  $F_{CA}$

|                            |  |       |       |       |       |
|----------------------------|--|-------|-------|-------|-------|
| $S_i S_{i-1}$<br>$S_{i+1}$ |  | 00    | 01    | 11    | 10    |
|                            |  | 0     | 1     | 1     | 0     |
| 0                          |  | $A_0$ | $A_1$ | $A_3$ | $A_2$ |
| 1                          |  | $A_4$ | $A_5$ | $A_7$ | $A_6$ |

$$Name = \sum_{i=0}^7 A_i 2^i$$

- Example:  $F_{CA,i} = S_{i-1} \oplus S_i$

|                            |  |    |    |    |    |
|----------------------------|--|----|----|----|----|
| $S_i S_{i-1}$<br>$S_{i+1}$ |  | 00 | 01 | 11 | 10 |
|                            |  | 0  | 1  | 1  | 0  |
| 0                          |  | 0  | 1  | 0  | 1  |
| 1                          |  | 0  | 1  | 0  | 1  |

Name =  $2^6 + 2^5 + 2^2 + 2^1 = 102$   
 This is called a **rule-102 cell**

# Quiz

Q1: Given a CA cell  $F_{CA,i} = S_{i-1} \oplus S_{i+1}$ . What is rule number?

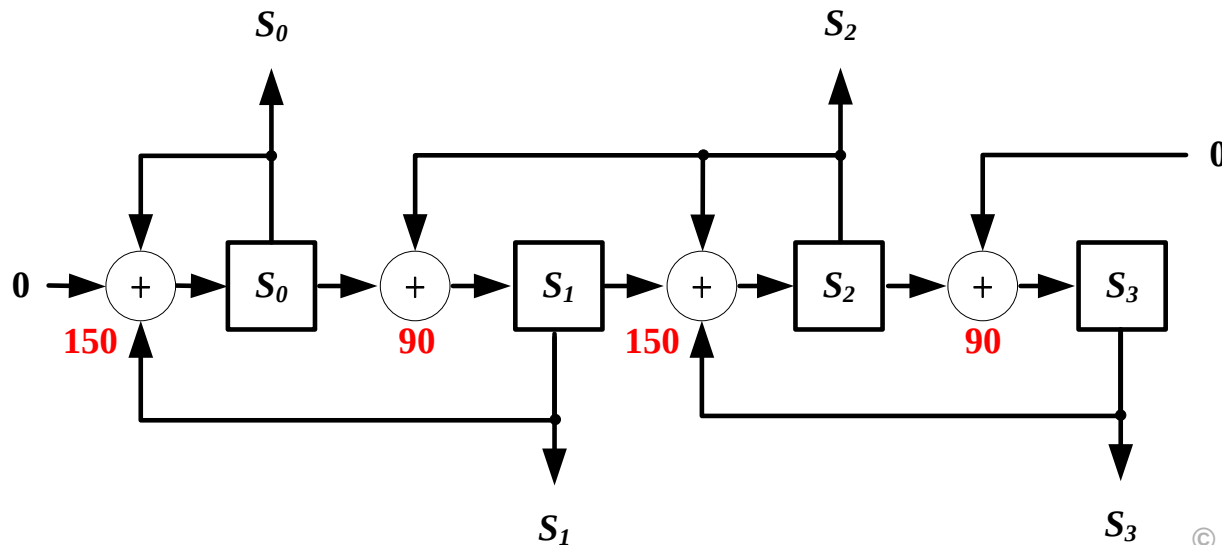
ANS:

K map of  $F_{CA}$

| $S_i S_{i-1}$ |   | 00    | 01    | 11    | 10    |
|---------------|---|-------|-------|-------|-------|
| $S_{i+1}$     | 0 | $A_0$ | $A_1$ | $A_3$ | $A_2$ |
|               | 1 | $A_4$ | $A_5$ | $A_7$ | $A_6$ |

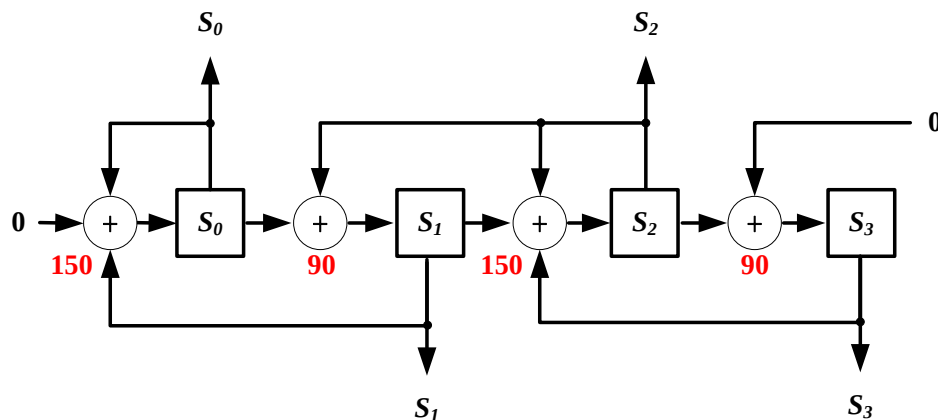
# Linear Hybrid CA, LHCA

- Rule-90 CA cell
  - ◆  $F_{CA,i} = S_{i-1} \oplus S_{i+1}$
- Rule-150 CA cell
  - ◆  $F_{CA,i} = S_{i-1} \oplus S_i \oplus S_{i+1}$
- **90/150 LHCA** with null boundary condition
  - ◆ Consist of only rule-90 and rule-150 cells
  - ◆ Very popular CA, math model well studied
- Example: 4-degree 90/150 LHCA



# State Transition

- Given seed = 1000
- CA generates pseudo random patterns
  - m-sequence of phase difference
  - Periodic
- Why this CA generates m-sequence?
  - to be proved later



| state | $S_0$ | $S_1$ | $S_2$ | $S_3$ |
|-------|-------|-------|-------|-------|
| 0     | 1     | 0     | 0     | 0     |
| 1     | 1     | 1     | 0     | 0     |
| 2     | 0     | 1     | 1     | 0     |
| 3     | 1     | 1     | 0     | 1     |
| 4     | 0     | 1     | 0     | 0     |
| 5     | 1     | 0     | 1     | 0     |
| 6     | 1     | 0     | 1     | 1     |
| 7     | 1     | 0     | 0     | 1     |
| 8     | 1     | 1     | 1     | 0     |
| 9     | 0     | 0     | 0     | 1     |
| 10    | 0     | 0     | 1     | 0     |
| 11    | 0     | 1     | 1     | 1     |
| 12    | 1     | 1     | 1     | 1     |
| 13    | 0     | 0     | 1     | 1     |
| 14    | 0     | 1     | 0     | 1     |
| 15    |       |       |       |       |

Phase diff. 11 9 1

# Quiz

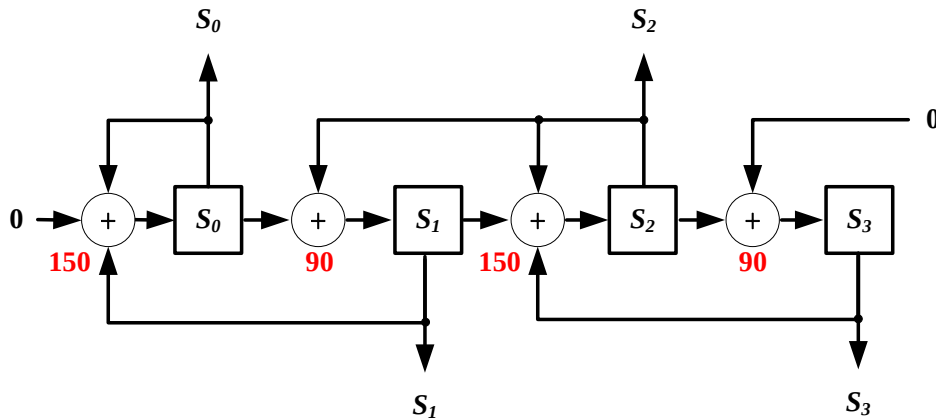
**Q1: What is state 15?**

**ANS:**

**Q2: What is cycle length of this CA?**

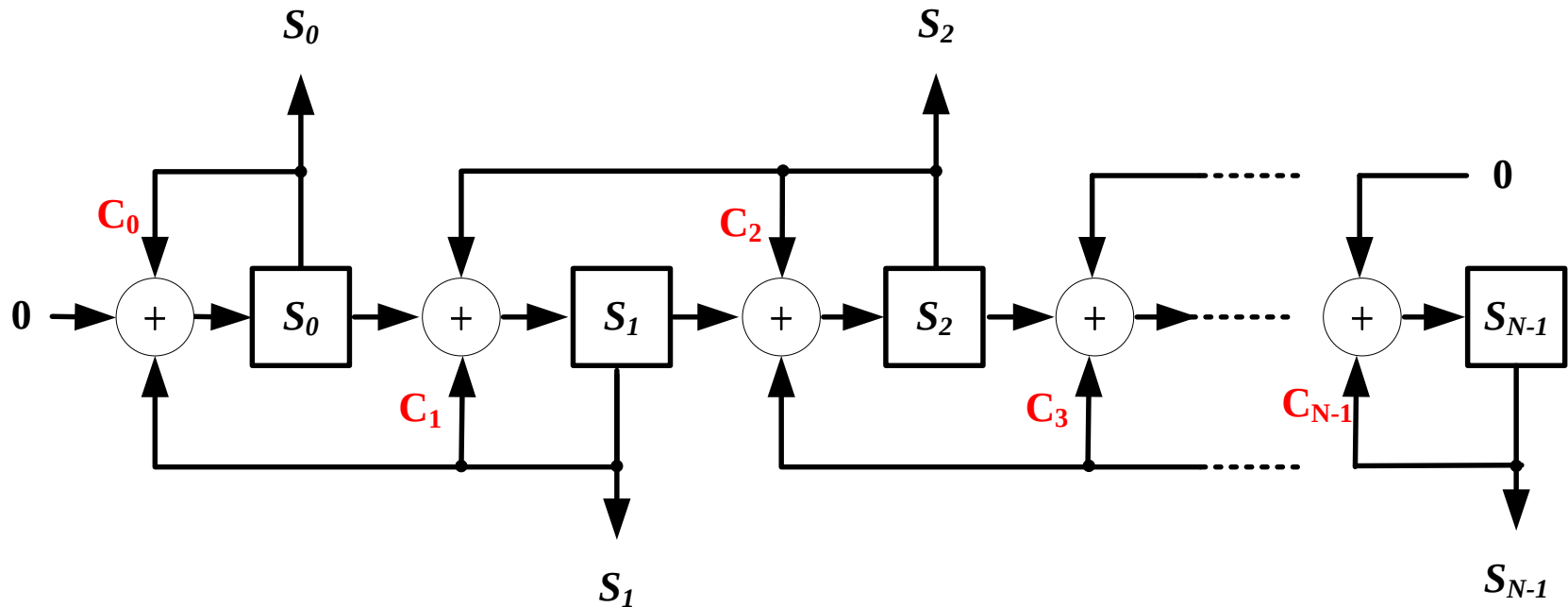
**ANS:**

| state | $S_0$ | $S_1$ | $S_2$ | $S_3$ |
|-------|-------|-------|-------|-------|
| 0     | 1     | 0     | 0     | 0     |
| 1     | 1     | 1     | 0     | 0     |
| 2     | 0     | 1     | 1     | 0     |
| 3     | 1     | 1     | 0     | 1     |
| 4     | 0     | 1     | 0     | 0     |
| 5     | 1     | 0     | 1     | 0     |
| 6     | 1     | 0     | 1     | 1     |
| 7     | 1     | 0     | 0     | 1     |
| 8     | 1     | 1     | 1     | 0     |
| 9     | 0     | 0     | 0     | 1     |
| 10    | 0     | 0     | 1     | 0     |
| 11    | 0     | 1     | 1     | 1     |
| 12    | 1     | 1     | 1     | 1     |
| 13    | 0     | 0     | 1     | 1     |
| 14    | 0     | 1     | 0     | 1     |
| 15    |       |       |       |       |



# N-degree 90/150 LHCA

- $C_i$  are coefficients
  - ♦ 1= feedback exists; **rule-150 cell**
  - ♦ 0 = no feedback; **rule-90 cell**



$$F_{CA,i} = S_{i-1} \oplus C_i S_i \oplus S_{i+1}$$



# Matrix Representation (90/150 LHCA)

$$\begin{bmatrix} S_0^+ \\ S_1^+ \\ \vdots \\ S_{N-3}^+ \\ S_{N-2}^+ \\ S_{N-1}^+ \end{bmatrix} = \begin{bmatrix} C_0 & 1 & 0 & \dots & 0 & 0 & 0 \\ 1 & C_1 & 1 & \dots & 0 & 0 & 0 \\ \vdots & 1 & C_2 & 1 & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & C_{n-3} & 1 & 0 \\ 0 & 0 & 0 & \dots & 1 & C_{n-2} & 1 \\ 0 & 0 & 0 & \dots & 0 & 1 & C_{n-1} \end{bmatrix} \begin{bmatrix} S_0 \\ S_1 \\ \vdots \\ S_{N-3} \\ S_{N-2} \\ S_{N-1} \end{bmatrix}$$

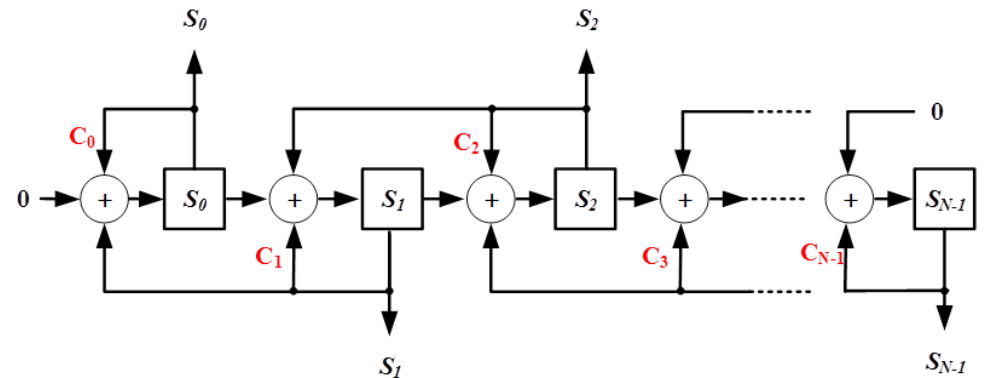
$S^+ = C S$  (mod-2 addition)

$S$  = current state

$S^+$  = next state

Characteristic polynomial:

$$f(\lambda) = \det (C - \lambda I)$$



# Quiz

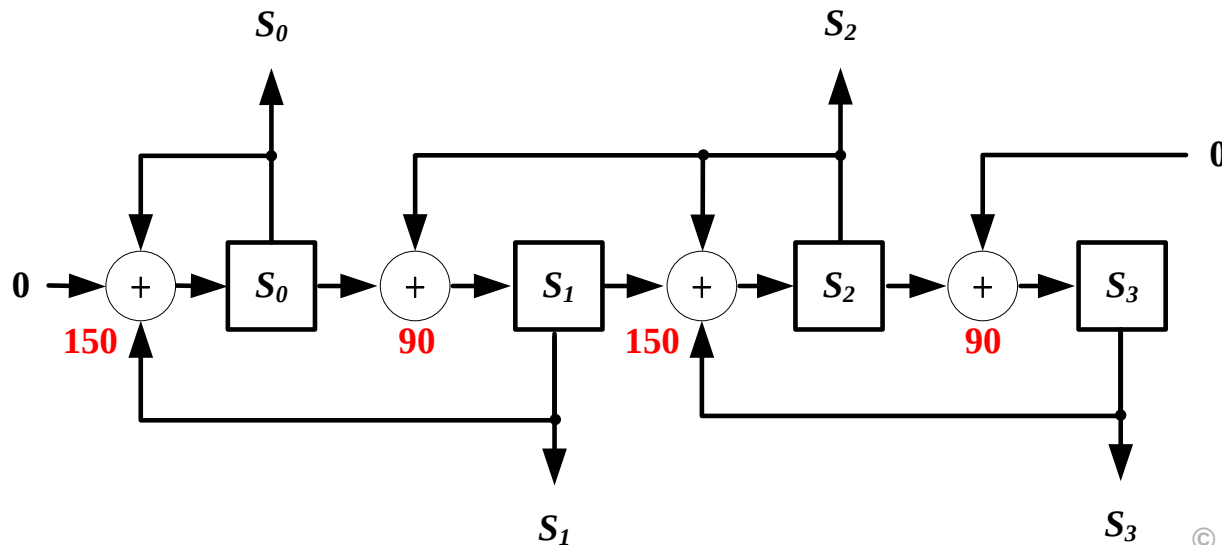
**Q1: Given this LHCA, what is matrix C ?**

**ANS:**

**Q2: What is characteristic polynomial?**

**NOTE : please use mod-2 addition**

**ANS:**



# Determinant

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{vmatrix} a & b & c & a & b \\ d & e & f & d & e \\ g & h & i & g & h \end{vmatrix}$$

$$\text{Determinant} = [aei + bfg + cdh] - [gce + hfa + idb]$$

# Determinant (column expansion)

- **Column expansion** (with respect to the  $j^{\text{th}}$  column)

$$\det(A) = |A| = \sum_{i=1}^n a_{ij} C_{ij} = a_{1j} C_{1j} + a_{2j} C_{2j} + \cdots + a_{nj} C_{nj}$$

- **Cofactor** of  $a_{ij} = C_{ij} = (-1)^{i+j} \times M_{ij}$ 
  - ♦  $M_{ij}$  is determinant of A by removing  $i^{\text{th}}$  row and  $j^{\text{th}}$  column

- Example:  $j=1$



$$\begin{aligned} \det(A) &= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} C_{11} + a_{21} C_{21} + a_{31} C_{31} = a_{11} (-1)^{1+1} M_{11} + a_{21} (-1)^{2+1} M_{21} + a_{31} (-1)^{3+1} M_{31} \\ &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{21} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{31} \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} \end{aligned}$$

# Characteristic Polynomial of CA

$$\det(C - \lambda I) = \det \begin{pmatrix} 1-\lambda & 1 & 0 & 0 \\ 1 & -\lambda & 1 & 0 \\ 0 & 1 & 1-\lambda & 1 \\ 0 & 0 & 1 & -\lambda \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$= (-1)^{1+1}(1-\lambda) \begin{vmatrix} -\lambda & 1 & 0 \\ 1 & 1-\lambda & 1 \\ 0 & 1 & -\lambda \end{vmatrix} + (-1)^{1+2}(1) \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1-\lambda & 1 \\ 0 & 1 & -\lambda \end{vmatrix}$$

$$= (1-\lambda)[\lambda^2(1-\lambda) + \lambda + \lambda] + (-1)[- \lambda(1-\lambda) - 1]$$

$$= \lambda^2(1-\lambda)^2 + 2\lambda(1-\lambda) + \lambda(1-\lambda) + 1$$

$$= \lambda^4 - 2\lambda^3 - 2\lambda^2 + 3\lambda + 1$$

$$= \lambda^4 + \lambda + 1 \quad (\text{mod-2 arithmetic})$$

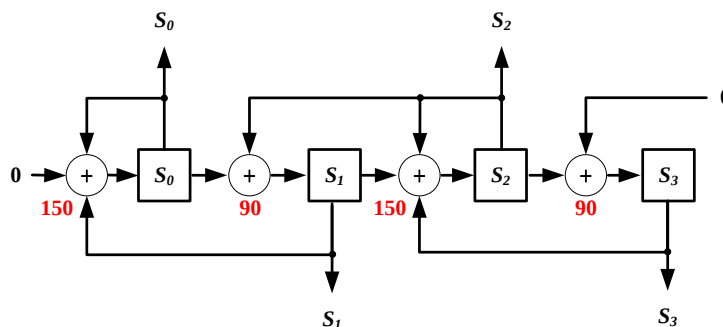
**This CA has a primitive polynomial**

# LFSR $\leftrightarrow$ CA \*not in exam

- If irreducible characteristic polynomial are equal:  $f_{CA}(x) = f_{LFSR}(x)$ 
  - ♦ Then their matrices are *similar*:  $T_{CA} \leftrightarrow T_{LFSR}$
  - ♦ i.e.  $T_{CA} = V^{-1}T_{LFSR}V$  (aka. **isomorphism transformation**)
  - ♦ [Serra 88][Bardell 90]

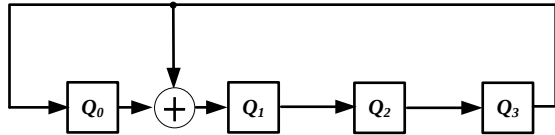
## Implications:

- 1. If characteristic polynomial  $f_{CA}(x)$  is **primitive polynomial**
  - ♦ It generates **m-sequence** of length  $2^N-1$
- 2. CA and LFSR of same polynomial generate same sequence
  - ♦ Just **phase difference**



$f_{CA}(x) = x^4 + x + 1$   
is primitive polynomial  
so it generates m-sequence

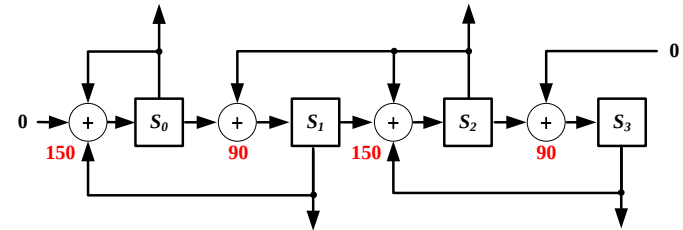
# LFSR $\leftrightarrow$ CA, same $f(x)=x^4+x+1$



| state | $Q_0$ | $Q_1$ | $Q_2$ | $Q_3$ |
|-------|-------|-------|-------|-------|
| 0     | 1     | 0     | 0     | 0     |
| 1     | 0     | 1     | 0     | 0     |
| 2     | 0     | 0     | 1     | 0     |
| 3     | 0     | 0     | 0     | 1     |
| 4     | 1     | 1     | 0     | 0     |
| 5     | 0     | 1     | 1     | 0     |
| 6     | 0     | 0     | 1     | 1     |
| 7     | 1     | 1     | 0     | 1     |
| 8     | 1     | 0     | 1     | 0     |
| 9     | 0     | 1     | 0     | 1     |
| 10    | 1     | 1     | 1     | 0     |
| 11    | 0     | 1     | 1     | 1     |
| 12    | 1     | 1     | 1     | 1     |
| 13    | 1     | 0     | 1     | 1     |
| 14    | 1     | 0     | 0     | 1     |

same  
m-sequence

Phase diff. 12 1 1



| state | $S_0$ | $S_1$ | $S_2$ | $S_3$ |
|-------|-------|-------|-------|-------|
| 0     | 1     | 0     | 0     | 0     |
| 1     | 1     | 1     | 0     | 0     |
| 2     | 0     | 1     | 1     | 0     |
| 3     | 1     | 1     | 0     | 1     |
| 4     | 0     | 1     | 0     | 0     |
| 5     | 1     | 0     | 1     | 0     |
| 6     | 1     | 0     | 1     | 1     |
| 7     | 1     | 0     | 0     | 1     |
| 8     | 1     | 1     | 1     | 0     |
| 9     | 0     | 0     | 0     | 1     |
| 10    | 0     | 0     | 1     | 0     |
| 11    | 0     | 1     | 1     | 1     |
| 12    | 1     | 1     | 1     | 1     |
| 13    | 0     | 0     | 1     | 1     |
| 14    | 0     | 1     | 0     | 1     |

same  
m-sequence

Phase diff. 11 9 1

# Pros and Cons of CA

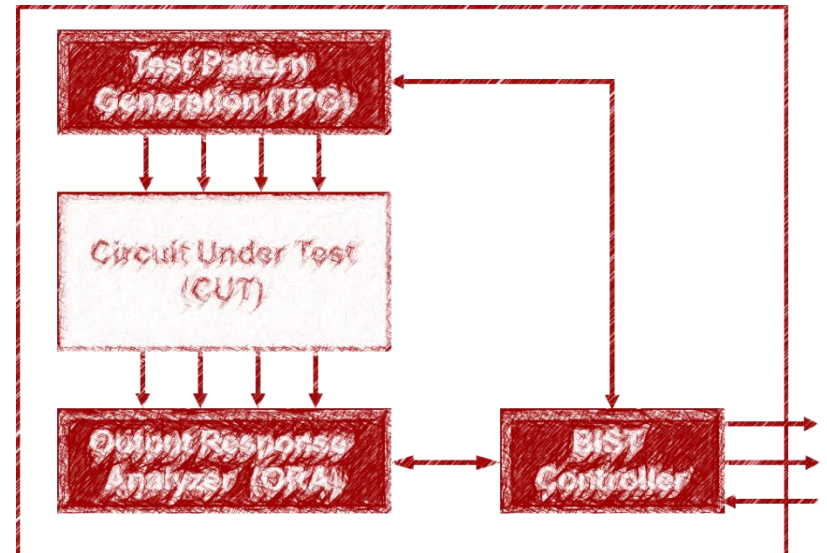
- Advantages of CA (over LFSR)
  - + **More randomness** than LFSR
    - because
      - ⇒ Large phase shift
  - + Potentially **faster** than LFSR
    - ⇒ short feedback loop
  - + **Small XOR**
    - ⇒ no more than 3-input XOR
- Disadvantages of CA (over LFSR)
  - **No easy design methodology**
    - ⇒ No good method to construct LHCA from polynomial





# Summary - CA

- CA are named by its **function**
- **90/150 LHCA** very popular
- Polynomial of CA is  $f(\lambda) = \det (C-\lambda I)$
- CA are good **but hard to design**



# Reference

- (BMS 87) P.H. Bardell, W.H. McAnney, J. Savior, *Built-in Test for VLSI: Pseudorandom Techniques*, Wiley Interscience, 1987.
- [Könemann 91] B Könemann, “LFSR-coded test patterns for scan designs,” European Test Conference, 1991